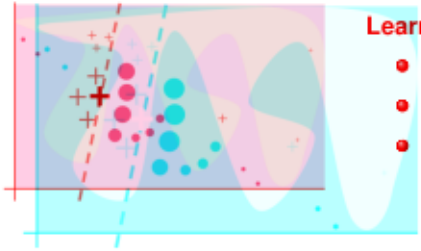
Interpretable Machine Learning

Introduction to Local Explanations



Learning goals

- Understand motivation for local explanations
- Develop an intuition for possible use-cases
- Know characteristics of local explanation methods

METHODOLOGICAL MOTIVATION

- Purpose of local explanations:
  - Insight into the driving factors for a **particular decision**
  - Understand the ML model's decisions in a **local neighborhood** of a given input
    (e.g., feature vector)

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  - Understand the ML model's decisions in a **local neighborhood** of a given input (e.g., feature vector)
- Local Methods can address questions such as:
  - **Why** did the model decide to predict $\hat{y}$ for input $\mathbf{x}$?
  - **How** does the model decide for observations that are similar to $\mathbf{x}$?
  - **What** would the ML model have decided if $\mathbf{x}$ its values in $\mathcal{X}$ were different?
  - **Where** (in which regions in $\mathcal{X}$) does the model fail?

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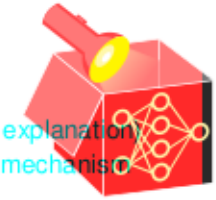
SOCIAL MOTIVATION



- Explanations for laypersons must be tailored to the **explainee** (who receives the explanation)
  $\rightsquigarrow$ **case specific**, **easy** for humans to understand, and **faithful** to the explained mechanism

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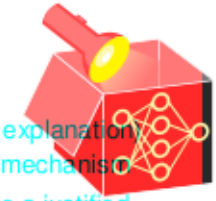

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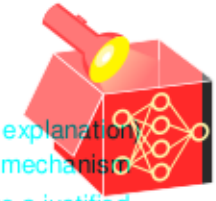

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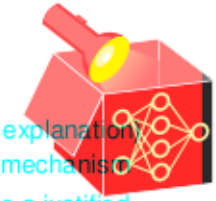

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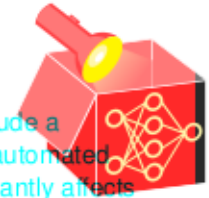
GDPR: THE RIGHT TO EXPLANATION



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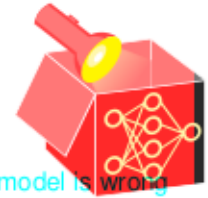
EXAMPLE: HUSKY OR WOLF?

- We trained a model to predict if an image shows a wolf or a husky
- Below the predictions on six test images are given
- Do you trust our predictor?

Source: [Sameer Singh 2018]



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EXAMPLE: HUSKY OR WOLF? USING LIME

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EXAMPLE: LOAN APPLICATION

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EXAMPLE: STOP OR RIGHT-OF-WAY?

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EXAMPLE: STOP OR RIGHT-OF-WAY?

- Imagine:
  - You work at a car company that develops image classifiers for autonomous driving
  - You show your model the following image (an adversarial example)
  - Classifier is 99% sure it describes a right-of-way sign
- Would you entrust other peoples lives into the hands of this software?

Source: [Eykholt et. al 2018]

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CHARACTERISTICS OF LOCAL EXPLANATIONS

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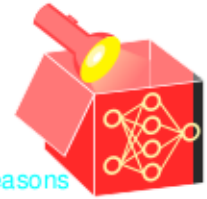

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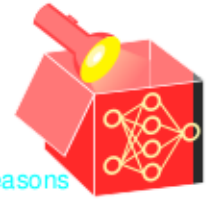

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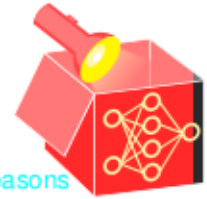

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- **Special:** Due to audience, strong interactions with social sciences and strong connections to cognitive science and neurosciences due to data types

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CREDIT DATASET

- We illustrate local explanation methods on the German credit data
- 522 complete observations, 9 features containing credit and customer information
- Binary target "risk" indicates if a customer has a 'good' or 'bad' credit risk
- We merged categories with few observations

name	type	range
age	numeric	[19, 75]
sex	factor	{male, female}
job	factor	{0, 1, 2, 3}
housing	factor	{free, own, rent}
saving.accounts	factor	{little, moderate, rich}
checking.accounts	factor	{little, moderate, rich}
credit.amount	numeric	[276, 18424]
duration	numeric	[6, 72]
purpose	numeric	{others, car, furniture, radio/TV}
risk	factor	{good, bad}